ModelIt!

Student Activity Pack

How Does Your Immune System Learn?

Modeling Vaccine Mechanisms and Herd Immunity Thresholds

Grade 11L1 | NGSS HS-LS1-2, HS-LS1-4 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **How do you think a vaccine protects you from getting sick — what is it actually doing inside your body?**

2. **Why do you think you need a new flu shot every year but your measles vaccine from childhood still works decades later?**

3. **If most people in a school are vaccinated against a disease, does that protect students who aren't vaccinated? Why or why not?**

4. **Draw a diagram showing what you think happens in your body between receiving a vaccine and being protected from the disease.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Vaccination Rate | Antigen Exposure | Antibody Production | Memory Cell Population | Herd Immunity Level | Disease Transmission Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Individual Immune Response

Single vaccination dose | 30-day antibody tracking + 6-month memory tracking

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Herd Immunity Buildup

Vaccination rate: 0% to 95% | Disease R0 = 5

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Outbreak in Under-Vaccinated Population

Vaccination rate: 80% | Disease requiring 93% immunity (measles)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Vaccine Efficacy Rate — The percentage of vaccinated individuals who develop full protective immunity — no vaccine is 100% effective; measles vaccine is 97%, flu vaccine ranges from 40-60%, COVID vaccines were 90-95% initially but declined against new variants

Variant Evolution — The tendency of pathogens to mutate, potentially producing variants that partially or fully escape vaccine-induced immunity — high transmission rates provide more opportunities for mutation, creating an evolutionary race between vaccination and viral adaptation

Vaccine Hesitancy — The spectrum of attitudes from full acceptance to full refusal of vaccination, influenced by trust in institutions, misinformation, historical medical abuse, and perceived risk-benefit calculations — hesitancy creates pockets of susceptibility that can sustain outbreaks

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does your immune system 'remember' a disease it has never had — and at what point does vaccinating enough people protect everyone, even those who can't be vaccinated? Use evidence from your simulation to support your answers.

1 How does your model illustrate the difference between individual immunity and population-level herd immunity?

2 Why is the herd immunity threshold a sharp transition rather than a gradual improvement — what does this tell you about how diseases spread?

3 How does your model demonstrate that people who cannot be vaccinated are directly harmed when vaccination rates drop below the herd immunity threshold?

4 What role does R0 play in determining how difficult a disease is to control through vaccination?

5 What are the limitations of your model in representing real-world vaccination challenges like hesitancy, access, and variant evolution?

8 STEM Challenge: Design a Vaccination Strategy for a New Pandemic

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YOUR ENGINEERING MISSION

Design a vaccination rollout strategy for a new respiratory virus with known R0 that achieves herd immunity in the shortest time while prioritizing the most vulnerable populations.

THE SCENARIO

A novel respiratory virus with an R0 of 4.5 has been identified, and a vaccine has just been approved. Your public health team must design a nationwide vaccination strategy. The herd immunity threshold is approximately 78%. You have limited initial vaccine supply (enough for 20% of the population per month), an immunocompromised population of 3% who cannot be vaccinated, and significant vaccine hesitancy in certain communities. Your plan must achieve herd immunity before the fall respiratory season.

Engineering Design Thinking

1. WHAT IS THE MATHEMATICALLY CALCULATED HERD IMMUNITY THRESHOLD FOR AN R0 OF 4.5, AND WHAT MARGIN OF ERROR SHOULD YOU BUILD IN?
2. HOW WILL YOU ADDRESS VACCINE HESITANCY — WHAT DOES YOUR MODEL SHOW ABOUT THE CONSEQUENCES OF SUB-THRESHOLD COVERAGE?
3. IF A NEW VARIANT EMERGES WITH R0 OF 7, HOW DOES YOUR STRATEGY NEED TO CHANGE?
4. WHAT MONITORING SYSTEM WILL YOU USE TO TRACK VACCINATION COVERAGE AND DETECT OUTBREAKS IN UNDER-VACCINATED POCKETS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Epidemiologists study how diseases spread through populations, designing vaccination strategies and outbreak response plans at the CDC, WHO, state health departments, and pharmaceutical companies, earning $75,000–$140,000/year. Immunologists research the molecular mechanisms of immune memory and vaccine development at universities, NIH, and biotech firms, earning $80,000–$160,000/year.